

YOLO project activity breakdown

Week	Task	SW	HW
Phase 1) Familiarization & Baseline			
1	Literature review (YOLO compression)	✓	
	Literature review (FPGA acceleration)		✓
2	SW Toolchain setup (e.g., TF, TF Lite, PyTorch) + YOLO baseline model selection	✓	
	HW Toolchain setup (e.g., PYNQ, Vivado, Vitis AI) + dataset selection		✓
3	Training and inference on an embedded platform (e.g., PYNQ ARM)	✓	✓
4	Profiling + bottleneck analysis (layer-wise latency, memory, FPS, CPU usage)	✓	✓
Phase 2) Co-design & optimization			
5	Model optimization: quantization + pruning exploration	✓	
	Parameterized YOLO layers simulation on FPGA		✓
6	Model optimization: layer-based error injection analysis	✓	
	Parameterized YOLO layers implementation on FPGA		✓
7	HW/SW partitioning (which layers to accelerate)	✓	✓
8	Model optimization based on quantization, pruning, and error injection analysis	✓	
	FPGA acceleration of the selected models using parameterized layers		✓
Phase 3) Evaluation			
9	Resource utilization + latency & throughput measurement	✓	✓
10	Comparison vs software baseline (accuracy vs speed trade-off)	✓	✓
11	Power & efficiency analysis + final evaluation	✓	✓
12	Paper drafts preparation	✓	✓